

TAIS Project Timeline

Telemetry Assessment and Integrity System | PAMCA698J - Dissertation-1 | Fall 2026-27 | 25MCA0022
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WK	DATES	GOAL / DELIVERABLE
1	Jul 24 - Jul 31	Literature survey write-up (expand refs into narrative). GitHub repo + Python env setup. Lock exact AIS-140 data schema.
2	Aug 1 - Aug 8	Write Problem Statement + Methodology docs for Review-1 (reuse existing architecture diagram). Start telemetry schema skeleton code.
3	Aug 9 - Aug 16	★ REVIEW-1 -- present problem statement, lit survey, architecture, methodology. Start real coding right after.
4	Aug 17 - Aug 23	Build the Telemetry Simulator -- highway / city / degraded-signal scenarios generating realistic AIS-140 fields.
5	Aug 24 - Aug 30	Build the Anomaly Injector -- all 6 types (position jump, drift, replay, GPS freeze, speed injection, heading mismatch).
6	Aug 31 - Sep 6	Build the Trust-Scoring Engine core -- implement Rules R1-R7 + the 100 minus Sum(penalties) formula.
7	Sep 7 - Sep 13	Add Confidence levels (High / Med / Low) and the Explanation Layer (human-readable reasons per flag).
8	Sep 14 - Sep 20	Build the Dashboard (Flask/Streamlit) -- vehicle list, score trend, alerts, tiered actions, matching the mockup.
9	Sep 21 - Sep 26	End-to-end integration testing (simulator to injector to engine to dashboard). Fix bugs. Buffer before Review-2.
10	Sep 27 - Oct 4	★ REVIEW-2 -- demo the working ~80% system live. (If pursuing the paper: only need to show 75% status.)
11	Oct 5 - Oct 11	Build the Evaluation Module -- precision / recall / F1 / latency against labeled simulator data. Tune rule weights.
12	Oct 12 - Oct 18	Fix issues evaluation exposes (esp. false positives on tunnel / low-HDOP cases). Start writing Final Dissertation Report.
13	Oct 19 - Oct 24	Finish both copies of Final Dissertation Report, run through Turnitin, get guide sign-off. Draft paper if pursuing it.
14	Oct 25 - Nov 3	★ GUIDE REVIEW -- submit 100%-complete project + 2 signed reports + Turnitin report.
15	Nov 4 - Nov 10	Build final presentation deck. Polish dashboard visuals for the external panel.
16	Nov 11 - Nov 15	Full mock run-through / rehearsal. Last fixes based on feedback.
17	Nov 16 - Nov 20	★ REVIEW-3 / FAT -- final presentation in front of external members.

★ = Fixed VIT review deadline (non-negotiable). Stretch goal: ML comparison layer -- cut this first if any phase runs long, never the rule engine or dashboard.

Today: 24 July 2026 -- Start Week 1 now.